Yi Cui

PhD Candidate in AI · Trustworthy Verification of Multimodal Reasoning & Agentic Systems

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Education

Ph.D. in Artificial Intelligence 2024 – Present
The Hong Kong University of Science and Technology (Guangzhou)
Supervisor: Prof. Hui Xiong — Founding Head of AI Thrust (HKUST-GZ); Fellow of ACM, IEEE, AAAS, AAAI, CAAI
Co-supervisor: Prof. Chengwei Qin
Passed the PhD Qualifying Examination (PQE), Aug. 2026

M.S. in Computer Engineering (GPA 3.5/4.0) 2024
New York University

B.S. in Computer Science (GPA 3.4/4.0; Minor in Economics) 2021
Pennsylvania State University

Research

My research builds toward trustworthy, process-level verification of multimodal and agentic systems. Starting from benchmark construction that exposes how i.i.d. evaluation overstates real-deployment reliability (SafeBuild-Bench, KDD'26), I moved to the verification layer itself: GLANCE predicts, from the original image alone, whether a vision–language model will follow an edit to the answer-relevant region, and my current process-reward-model work asks whether step scores track the image facts that decide each step. I am also studying what it takes for pretrained video world models to obey game rules, and expanding my PQE survey, from visual chain-of-thought to verifiable visual-thinking agents, into an ACM Computing Surveys submission.

Publications

† *Co-first author* ∗ *Corresponding author*

[1] **SafeBuild-Bench: A Temporal-Robust Construction Safety Benchmark with Graph-Enhanced Data Mining.**
Yi Cui†, Zilin Wang†, Yijie Xu, Qianyi Cai, Huizai Yao, Shuai Jiang, Bingzhuo Zhong∗, Hui Xiong∗. **ACM SIGKDD 2026 (KDD)**, Datasets & Benchmarks Track. arXiv:2608.00068

[2] **Beyond Boundaries: Leveraging Vision Foundation Models for Source-Free Object Detection.**
Huizai Yao, Sicheng Zhao, Pengteng Li, **Yi Cui**, Shuo Lu, Weiyu Guo, Yunfan Lu, Yijie Xu, Hui Xiong∗. **AAAI 2026**. arXiv:2511.07301

[3] **From Events to Enhancement: A Survey on Event-Based Imaging Technologies.**
Yunfan Lu, Xiaogang Xu, Pengteng Li, Yusheng Wang, **Yi Cui**, Huizai Yao, Hui Xiong. **arXiv** preprint, 2025. arXiv:2505.05488

Manuscripts under Review

[4] **Change the Image, Change the Answer? Predicting VLM Responses to Regional Edits.**
Yi Cui, Yili Wang, Zilin Wang, Huizai Yao, Chengwei Qin, Hui Xiong. Under review, **ICLR 2027**.

When a VLM answers an image question correctly, can measurements on the original image predict whether it will follow an edit to the answer-relevant region? GLANCE, a training-free three-view score (full image, enlarged crop, masked region) that never sees the edited image, ranks edit-following above preservation at AUROC 0.71–0.83 across six Visual7W model cohorts and beats the strongest attention and geometry readouts in five.

Ongoing Research

- **Multimodal process reward models** (targeting **ACL 2027**). A paired-image diagnostic benchmark testing whether PRM step scores follow the image facts that decide each step: the step's text stays fixed while an image edit flips its truth. It builds on an audit of six multimodal PRMs in which step ranking and first-error detection disagree.

- **Game world models.** Using a program harness so that pretrained video models generate game video that obeys game mechanics, and measuring the cost: how much input detail, training data, and verification an event needs when it runs against the model's visual prior (Sokoban, Griddly, Doom, Hollow Knight).
- **Survey.** Expanding my PQE survey, *From Visual Chain-of-Thought to Verifiable Visual-Thinking Agents*, into an ACM Computing Surveys submission on verifying the intermediate representations of visual reasoning.

Projects

Construction-Site Safety-Hazard VLM (Guangdong Heavy Industry, Industry Collaboration) 2025 – Present

Qwen3.6-27B · Full-parameter SFT + LoRA (ms-swift) · vLLM FP8 serving · Ascend NPU deployment · Expert data audits

- Lead modeling, data, and evaluation for a VLM that flags safety hazards in construction-site inspection photos; the model is deployed for the industry partner, on a single-GPU FP8 server and on the partner's Ascend NPU service.
- Fine-tuned Qwen3.6-27B on 100K+ inspection images labeled under an expert-revised 231-class hazard taxonomy. Across all 1,156 independent test scenes, strict top-1 accuracy rose from 24.9% (base model) to 45.9%, while a Qwen3-VL-235B teacher reached only 27.4% — the task hinges on the partner's own labeling conventions, so I dropped the planned teacher distillation.
- Traced a weak top-3 score to the training data (only 0.15% multi-label samples, so the model reported one hazard per image) and fixed it in the prompt rather than the training target: requiring exactly three candidates raised top-3 accuracy from 38.3% to 58.1% with no retraining.
- Rebuilt the evaluation for deployment validity: sampled by inspection scene instead of by image (10,173 test images were only 1,156 scenes), removed test photos duplicated in training via content hashing (∼10%), and ran two rounds of expert audits that put image–description mismatch at 10–19%, so every reported accuracy names its subset.
- Served the model with online FP8 quantization in vLLM on one NVIDIA L20 with no measurable accuracy loss (top-3 65.8% vs. 64.6% in BF16, within noise), and checked the partner's Ascend deployment against internal evaluation (top-3 60.0% vs. 60.1% on 750 validation images).

RedDust — Long-Horizon LLM-Agent Benchmark (open source: github.com/peter-cui-yi/RedDust) 2026

TypeScript deterministic engine · React/Phaser replay · LLM-as-judge with human agreement checks · Six LLM families

- Built a 30-day survival-shelter benchmark that scores not just whether an LLM agent survives but whether it stays accountable: each run has 110 decisions, a ledger tracks the agent's Day-0 promises, and scores form a non-compensatory profile (outcome, integrity, durability, drift, comprehension).
- In a preliminary cross-model panel, agents with matched short-horizon social skill still split on long-horizon consistency: those that collapse keep their stated commitments but let the shelter decline, and the same planning scaffold or commitment reminder rescues some models while breaking others.

Skills

Research Areas: Vision-Language Models (VLM/MLLM) · Multimodal Evaluation & Benchmarking · Process Reward Models (PRM) / Verifiable Reasoning · LLM-Agent Evaluation · Data-Centric AI

Training & Deployment: Full-parameter SFT · LoRA / PEFT · Knowledge Distillation · Contrastive Learning · Distributed Training (multi-GPU) · FP8 Inference Serving

Programming: Python · C++ · Java · TypeScript **Frameworks/Tools:** PyTorch · ms-swift · LLaMA-Factory · vLLM · React/Phaser · MATLAB

Teaching

Teaching Assistant, HKUST (Guangzhou)

AIAA 3053 — Reinforcement Learning: Principles and Methods UFUG 2104 — Applied Statistics

Languages

English (Fluent — 6 years of study in the U.S.) Chinese (Mandarin, Native)